

Improved recovery time in single coil pulsed NMR spectrometers

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Improved recovery time in single coil pulsed NMR spectrometers

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A simple modification to a conventional single coil pulsed NMR spectrometer is described which damps the final stage of the transmitter at low rf levels. For an unmodified Bruker SXP spectrometer a receiver recovery time of 7.7 μsec is obtained at 60 MHz; this time improves to 5.5 μsec on adding the "low level damper."

Pulsed NMR studies in the solid state require observation of the nuclear signal only microseconds after the spin system is perturbed by an intense rf pulse. In a single coil spectrometer the transmitter and receiver share the sample coil, and therefore no NMR signal can be seen during the time that the sample coil rings down from the kilovolt level of the transmitter pulse to the millivolt level of the induced nuclear signal. To minimize receiver recovery time, the energy of the transmitter pulse must be rapidly dissipated after the transmitter is turned off. The literature and oral folklore are rich with fast recovery probe schemes; representative ones are discussed elsewhere.¹⁻⁴

We find that conventional single coil schemes do not properly damp the transmitter pulse at rf levels below a volt. A simple modification, to be described, provides damping at this low level and significantly improves the receiver recovery time.

A somewhat idealized single coil system is shown in Fig. 1. The transmitter, receiver preamplifier, and probe circuit are matched to 50 Ω , and 50- Ω coaxial cable is used for the half-wave ($\lambda/2$) and quarter-wave ($\lambda/4$) lines. During the transmitter pulse, diodes D1 connect the probe to the transmitter. The rf pulse also turns on diodes D2, protecting the receiver preamplifier against severe overload. These latter diodes are separated from the probe by a $\lambda/4$ cable and so do not affect the probe circuit. (The cable from the transmitter to the probe circuit need not be exactly $\lambda/2$ long pro-

vided the probe circuit does indeed present a 50- Ω impedance.)

The weakness of this system is seen as the rf pulse is turned off. As the voltage across diodes D1 drops below about one volt, their impedance increases, and they tend to disconnect the probe from the transmitter. Note that the final stage of an NMR transmitter is generally a tuned output transformer, as shown in Fig. 1. If diodes D1 truly disconnect the probe, then the tuned circuits of the transmitter and probe would ring down independently. The transmitter would then ring down much more slowly, since it would no longer see the load of the probe.

Unfortunately, in practice the probe and transmitter are still coupled, albeit weakly, by the capacitance C (approx. 2 pF per diode) of diodes D1. This coupling ties the voltage across the sample coil to the slow ringdown on the transmitter's final stage, and so the receiver recovery time is prolonged.

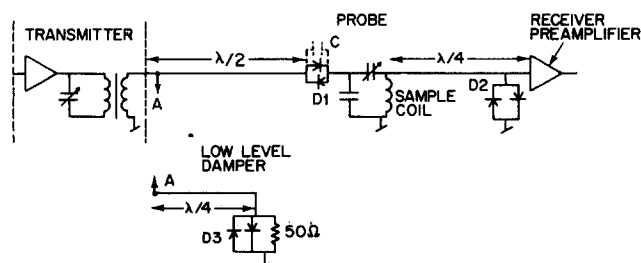


FIG. 1. The elements of a single coil spectrometer. Also shown is a low level damper (connected at point A) consisting of diodes D3, a 50- Ω load, and a $\lambda/4$ line. All diodes are 1N4448; D1 and D2 each use three pairs, while D3 consists of one pair.

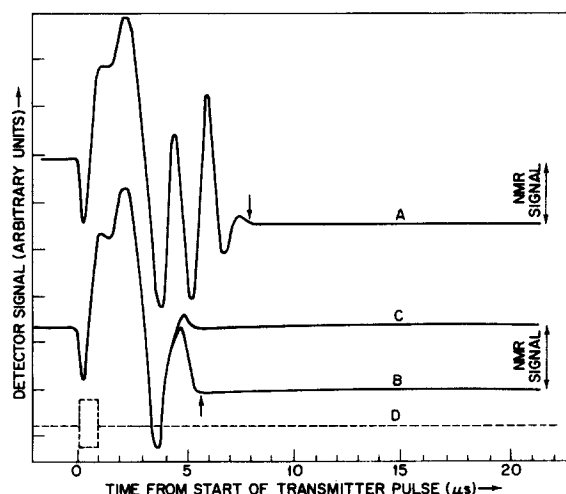


FIG. 2. Detector signals following a 0.8- μsec $\pi/2$ rf pulse. Curve A is for the unmodified spectrometer. The low level damper has been used in curves B and C, which show the receiver response on and well off resonance, respectively. (Curves B and C have been shifted vertically for clarity.) The arrows indicate when the receiver has recovered, at 7.7 μsec in curve A, and 5.5 μsec in curve B. Curve D indicates the timing of the transmitter pulse.

This problem is overcome by connecting to the transmitter a device which has an "appetite" only for low level rf energy. Such a modification—a "low level damper"—is shown in Fig. 1. For high rf levels (above one volt), diodes D3 conduct strongly. At the transmitter end of the $\lambda/4$ line this appears as a high impedance. At levels below a volt, the diodes D3 turn off, and the transmitter sees the 50- Ω load of the damper. Hence, the transmitter has somewhere to dump its energy even at low power levels. Since the damper presents a high impedance during the transmitter pulse, no sacrifice of rf magnetic field strength in the sample coil is required. The low level damper circuit may be connected anywhere along the $\lambda/2$ cable of the transmitter. (In the fast recovery delay line probe developed by Lowe and Engelsberg,⁴ a similar low level damper was used, not to improve recovery time, but rather to reduce the standing wave ratio for the induced low level NMR signal.)

We have tested such a low level damper on a Bruker SXP spectrometer. The system, operating at 60 MHz, is substantially as indicated in Fig. 1. The receiver recovery time was measured from the *start* of a $\pi/2$ rf pulse to the point when the receiver output has reached 95% of the full NMR signal. In these tests a field strength of 75 G was used, corresponding to a $\pi/2$ pulse of 0.8 μ sec for protons. The proton resonance of 0.25 ml of glycerol was observed, using a receiver bandwidth of approximately 1 MHz.

Figure 2 shows the signal from the phase-sensitive detector of the receiver, recorded by a Biomation 610 analog-to-digital converter. In the figure curve A shows a receiver recovery time of 7.7 μ sec for the unmodified spectrometer. (Note that the NMR signal is negative.) Curves B and C, displaced vertically for clarity, show the results using the low level damper; curve B is taken on resonance, while in curve C the static magnetic field has been shifted well off resonance, and no NMR signal is seen. The low level damper has reduced the receiver recovery time to 5.5 μ sec. We attribute the improvement to the damping of the final tuned stage of the transmitter, at rf levels below one volt.

It must be stressed that other mechanisms, such as the recovery characteristics of the receiver system from overload, influence the receiver recovery time, and so the low level damper is no panacea.

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